

Advanced Machine Learning

DATA 642

Lecture 2 — Second order optimization

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

Useful videos: https://www.youtube.com/watch?v=W7S94pq5Xuo&t=4s&ab_channel=VisuallyExplained

In the realm of unconstrained optimization, second-order optimization methods offer superior techniques by approximating the objective function with a function that is easy to optimize. One such method is Newton's method, which iteratively optimizes a quadratic approximation of the objective function.

Newton's Method

Newton's method utilizes the second-order Taylor series expansion of the objective function $J(\boldsymbol{\theta})$ around a current point $\boldsymbol{\theta}^{(k)}$:

$$J(\boldsymbol{\theta}) \approx J(\boldsymbol{\theta}^{(k)}) + \boldsymbol{\delta}^\top \nabla f(\boldsymbol{\theta}^{(k)}) + \frac{1}{2} \boldsymbol{\delta}^\top H(\boldsymbol{\theta}^{(k)}) \boldsymbol{\delta} \quad (1)$$

Here, $H(\boldsymbol{\theta}^{(k)}) = \nabla^2 J(\boldsymbol{\theta}^{(k)})$ represents the Hessian matrix.

Newton's method iteratively solves the equation $H(\boldsymbol{\theta}^{(k)}) \boldsymbol{\delta} = -\nabla J(\boldsymbol{\theta}^{(k)})$ to find the update direction $\boldsymbol{\delta}$, then updates the parameters as $\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \boldsymbol{\delta}$.

Limitations of Newton's Method

Newton's method may diverge if the Hessian matrix is not positive definite. One solution is to add a multiple of the identity matrix to the Hessian to ensure positive definiteness, leading to the Levenberg-Marquardt algorithm.

Challenges with Analytical Derivatives

Analytically determining second derivatives for the Hessian matrix can be challenging. Finite difference approximations offer an alternative, but they may not guarantee positive definiteness.

Quasi-Newton Methods

Quasi-Newton methods approximate the inverse of the Hessian matrix with a symmetric positive definite matrix that is updated iteratively. Basic steps include:

- Setting $\delta^{(k)} = -\mathbf{H}_*(\boldsymbol{\theta}^{(k)})\nabla J(\boldsymbol{\theta}^{(k)})$.
- Conducting line search along $\delta^{(k)}$ to find $\boldsymbol{\theta}^{(k+1)} = \boldsymbol{\theta}^{(k)} + \alpha^{(k)}\delta^{(k)}$.
- Updating $\mathbf{H}_*^{(k)}$ to yield $\mathbf{H}_*^{(k+1)}$.

Quasi-Newton methods offer efficient and stable optimization techniques without the need for analytical second derivatives.

BFGS Algorithm

Another widely used quasi-Newton method is the Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm. It updates an approximation of the inverse Hessian matrix using rank-two updates, making it computationally efficient and robust for large-scale optimization problems.

- The BFGS algorithm iteratively refines the approximation of the Hessian matrix by updating a symmetric positive definite matrix that approximates the inverse Hessian. It does so using the information from the gradient and parameter updates at each iteration. The BFGS update formula involves rank-two updates to the approximation matrix, ensuring that it remains positive definite throughout the optimization process.
- Compared to Newton's method, the BFGS algorithm offers several advantages. It does not require the computation of second derivatives explicitly, making it suitable for problems where analytical derivatives are not available or expensive to compute. Additionally, the BFGS algorithm tends to exhibit faster convergence rates and better stability, especially for high-dimensional optimization problems.